



Chapter 3 Fundamentals of Fluid Motion

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Chapter 3 Fundamentals of Fluid Motion

Section 4



Basic equations of fluid flow

- 1, Continuity equation
- 2, Momentum balance for ideal fluid
- 3, Momentum balance for real fluid

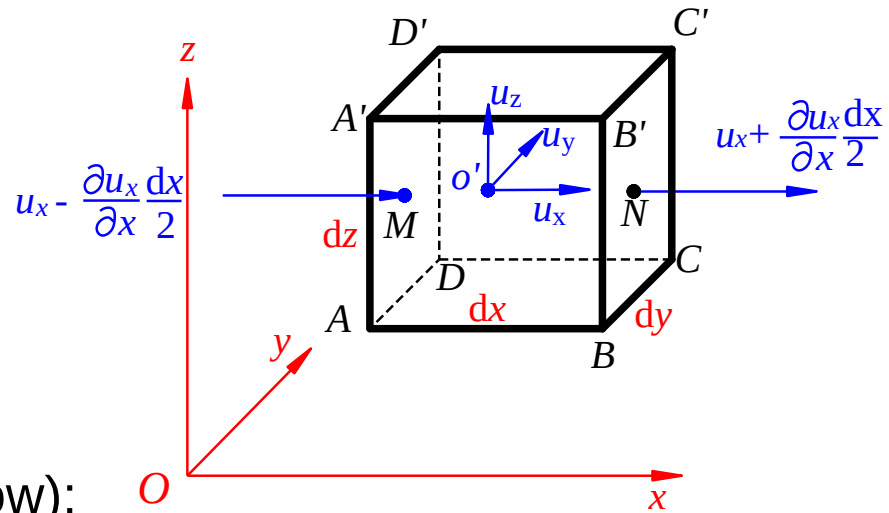


Section 4 Basic equations of fluid flow

1, Continuity equation

$$(\rho u)_M = \rho u_x - \frac{1}{2} \frac{\partial(\rho u_x)}{\partial x} dx$$

$$(\rho u)_N = \rho u_x + \frac{1}{2} \frac{\partial(\rho u_x)}{\partial x} dx$$



Mass loss (outflow minus mass inflow):

$$\begin{aligned} \Delta M_x &= M_{\text{右}} - M_{\text{左}} = \left[\rho u_x + \frac{1}{2} \frac{\partial(\rho u_x)}{\partial x} dx \right] dy dz - \left[\rho u_x - \frac{1}{2} \frac{\partial(\rho u_x)}{\partial x} dx \right] dy dz \\ &= \frac{\partial(\rho u_x)}{\partial x} dx dy dz \end{aligned}$$



$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_x)}{\partial x} + \frac{\partial(\rho u_y)}{\partial y} + \frac{\partial(\rho u_z)}{\partial z} = 0$$

(1) Steady flow of compressible fluid: $\frac{\partial \rho}{\partial t} = 0$

有
$$\frac{\partial(\rho u_x)}{\partial x} + \frac{\partial(\rho u_y)}{\partial y} + \frac{\partial(\rho u_z)}{\partial z} = 0$$

(2) Incompressible fluid: $\rho = C$

有
$$\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} = 0$$



Example 1: 2D flow with (1) $u_x = -2y$, $u_y = 3x$ or (2) $u_x = 0$, $u_y = 3xy$, $\rho = \text{const}$, are they meaningful?

$$(1) \quad \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} = \frac{\partial(-2y)}{\partial x} + \frac{\partial(3x)}{\partial y} = 0 + 0 = 0$$

Obeys continuity equation of incompressible fluid.

Meaningful.

$$(2) \quad \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} = \frac{\partial(0)}{\partial x} + \frac{\partial(3xy)}{\partial y} = 0 + 3x \neq 0$$

$\rho = \text{const}$ implies an incompressible fluid. However, continuity equation of incompressible fluids is not obeyed. Not meaningful.



Example 2:

(1) Is this flow meaningful? $u_x = \frac{1}{\rho}t, u_y = \frac{1}{\rho}3xy, u_z = \frac{1}{\rho}xz, \rho = t$

(2) If you leave u_x, u_y, ρ unchanged, how would you have to alter u_z ?

$$(1) \quad \frac{\partial \rho}{\partial t} + \frac{\partial \rho u_x}{\partial x} + \frac{\partial \rho u_y}{\partial y} + \frac{\partial \rho u_z}{\partial z} = 1 + 0 + 3x + x \neq 0$$

Does not obey continuity equation. Not meaningful.

$$(2) \quad \frac{\partial \rho u_z}{\partial z} = -(1 + 0 + 3x) = -1 - 3x$$

$$u_z = \frac{-z}{\rho}(1 + 3x) + \frac{1}{\rho}f(x, y)$$

$$u_z = \frac{-z}{\rho}(1 + 3x)$$





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Basic equations of fluid flow

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2, Momentum balance for ideal fluid (Euler equations)

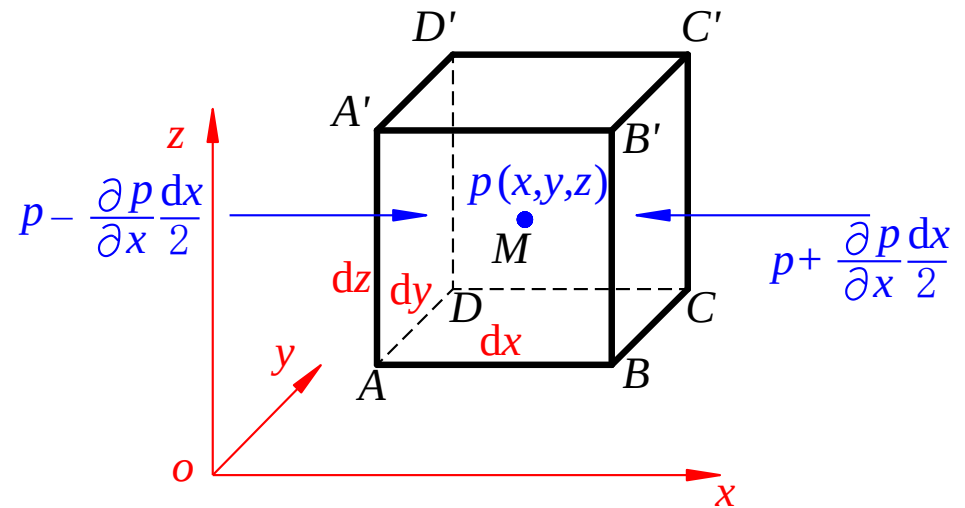
Surface forces

$$\left\{ \begin{array}{l} (p - \frac{\partial p}{\partial x} \frac{dx}{2}) dydz \\ (p + \frac{\partial p}{\partial x} \frac{dx}{2}) dydz \end{array} \right.$$

Mass force: $f_x \rho dx dy dz$

Newton:

$$\sum F = ma :$$



$$(p - \frac{\partial p}{\partial x} \frac{dx}{2}) dydz - (p + \frac{\partial p}{\partial x} \frac{dx}{2}) dydz + f_x \rho dx dy dz = \rho dx dy dz \frac{du_x}{dt}$$

$$f_x - \frac{1}{\rho} \frac{\partial p}{\partial x} = \frac{du_x}{dt} = \frac{\partial u_x}{\partial t} + u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} + u_z \frac{\partial u_x}{\partial z}$$





$$\left. \begin{aligned} f_x - \frac{1}{\rho} \frac{\partial p}{\partial x} &= \frac{du_x}{dt} \\ f_y - \frac{1}{\rho} \frac{\partial p}{\partial y} &= \frac{du_y}{dt} \\ f_z - \frac{1}{\rho} \frac{\partial p}{\partial z} &= \frac{du_z}{dt} \end{aligned} \right\}$$

If $\frac{du_x}{dt}, \frac{du_y}{dt}, \frac{du_z}{dt}$ vanish, these equations reduce to the equations of stress equilibrium.

General momentum balance of ideal fluid: $\vec{f} - \frac{1}{\rho} \nabla p = \frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla) \vec{u}$
(Euler equations of motion)





Chapter 3 Fundamentals of Fluid Motion

Section 4 Basic equations of fluid flow

- 1, Continuity equation
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3, Momentum balance for real fluid

1, Characteristics of real fluids

- (1) Surface forces include compressive stress and shear stress, determined by the generalized Newton's stress formula:

$$\left. \begin{aligned} \tau_{xy} &= \mu \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) = \tau_{yx} \\ \tau_{yz} &= \mu \left(\frac{\partial u_y}{\partial z} + \frac{\partial u_z}{\partial y} \right) = \tau_{zy} \\ \tau_{zx} &= \mu \left(\frac{\partial u_z}{\partial x} + \frac{\partial u_x}{\partial z} \right) = \tau_{xz} \end{aligned} \right\}$$

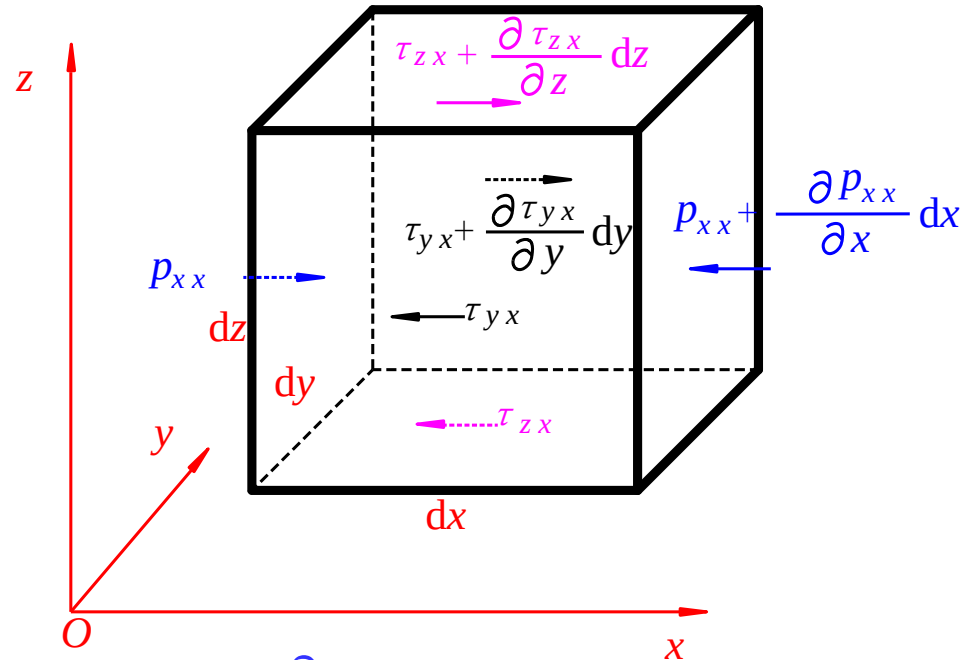
- (2) Due to viscous stress, compressive stresses are no longer isotropic, $p_{xx} \neq p_{yy} \neq p_{zz}$ and $p = \frac{1}{3}(p_{xx} + p_{yy} + p_{zz})$.

For incompressible flows, the pressure components are given by (not shown):

$$\left. \begin{aligned} p_{xx} &= p - 2\mu \frac{\partial u_x}{\partial x} \\ p_{yy} &= p - 2\mu \frac{\partial u_y}{\partial y} \\ p_{zz} &= p - 2\mu \frac{\partial u_z}{\partial z} \end{aligned} \right\}$$



2, Balance equations of real fluids



x-direction:

$$\sum F = ma \quad :$$

$$\begin{aligned} & f_x \rho dx dy dz + [p_{xx} dy dz - (p_{xx} + \frac{\partial p_{xx}}{\partial x} dx) dy dz] \\ & - [\tau_{yx} dx dz - (\tau_{yx} + \frac{\partial \tau_{yx}}{\partial y} dy) dx dz] \\ & - [\tau_{zx} dy dx - (\tau_{zx} + \frac{\partial \tau_{zx}}{\partial z} dz) dy dx] \\ & = \rho dx dy dz \frac{du_x}{dt} \end{aligned}$$



Incompressible flow: $\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} = 0$

- leads to incompressible Navier-Stokes (N-S) equations:

$$\left. \begin{aligned} f_x - \frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \nabla^2 u_x &= \frac{du_x}{dt} \\ f_y - \frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \nabla^2 u_y &= \frac{du_y}{dt} \\ f_z - \frac{1}{\rho} \frac{\partial p}{\partial z} + \nu \nabla^2 u_z &= \frac{du_z}{dt} \end{aligned} \right\}$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \quad \nabla^2 u_x = \frac{\partial^2 u_x}{\partial x^2} + \frac{\partial^2 u_x}{\partial y^2} + \frac{\partial^2 u_x}{\partial z^2}$$

$$\vec{f} - \frac{1}{\rho} \nabla p + \nu \nabla^2 \vec{u} = \frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla) \vec{u}$$





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Chapter 3 Fundamentals of Fluid Motion

Section 5 Integration of Euler equations of motion



1, Integration for steady, potential,
incompressible flow under only gravity

2, Path integration along streamline for steady,
incompressible flow under only gravity





Integration of Euler equations not possible for general situation, but only for certain conditions.

$$\left. \begin{aligned} f_x - \frac{1}{\rho} \frac{\partial p}{\partial x} &= \frac{du_x}{dt} \\ f_y - \frac{1}{\rho} \frac{\partial p}{\partial y} &= \frac{du_y}{dt} \\ f_z - \frac{1}{\rho} \frac{\partial p}{\partial z} &= \frac{du_z}{dt} \end{aligned} \right\}$$

The Euler equations are multiplied by **dx**, **dy**, and **dz**, respectively, and added:

$$(f_x dx + f_y dy + f_z dz) - \frac{1}{\rho} \left(\frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz \right) = \frac{du_x}{dt} dx + \frac{du_y}{dt} dy + \frac{du_z}{dt} dz$$



1, Integration for potential flow

$$1, \text{ steady: } \frac{\partial p}{\partial t} = 0, \text{ incompressible: } \rho = \mathbf{C} \quad \langle \mathbf{II} \rangle = \frac{1}{\rho} dp = d\left(\frac{p}{\rho}\right)$$

2, only gravity : $f_x=f_y=0$, $f_z = -g$ $\langle I \rangle = -gdz$

3, potential flow: $\frac{\partial u_x}{\partial y} = \frac{\partial u_y}{\partial x}$, $\frac{\partial u_y}{\partial z} = \frac{\partial u_z}{\partial y}$, $\frac{\partial u_x}{\partial z} = \frac{\partial u_z}{\partial x}$

$$\frac{du_x}{dt} dx \longrightarrow (u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_y}{\partial x} + u_z \frac{\partial u_z}{\partial x}) dx$$

$$= \frac{\partial}{\partial x} \left[\frac{1}{2} (u_x^2 + u_y^2 + u_z^2) \right] dx = \frac{\partial}{\partial x} \left(\frac{u^2}{2} \right) dx$$

$$\langle \text{III} \rangle = \frac{\partial}{\partial x} \left(\frac{u^2}{2} \right) dx + \frac{\partial}{\partial y} \left(\frac{u^2}{2} \right) dy + \frac{\partial}{\partial z} \left(\frac{u^2}{2} \right) dz = d \left(\frac{u^2}{2} \right)$$





$$(f_x dx + f_y dy + f_z dz) - \frac{1}{\rho} \left(\frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz \right) = \frac{du_x}{dt} dx + \frac{du_y}{dt} dy + \frac{du_z}{dt} dz$$

<I> <II> <III>

1, Integration for potential flow

$$< I > = -g dz$$

$$< II > = -d \frac{p}{\rho}$$

$$< III > = d \left(\frac{u^2}{2} \right)$$

Hence,
$$-g dz - d \left(\frac{p}{\rho} \right) = d \left(\frac{u^2}{2} \right)$$

$$z + \frac{p}{\rho g} + \frac{u^2}{2g} = C$$



- Ideal potential stream, Bernoulli's equation

$$z + \frac{p}{\rho g} + \frac{u^2}{2g} = C$$

$$z_1 + \frac{p_1}{\rho g} + \frac{u_1^2}{2g} = z_2 + \frac{p_2}{\rho g} + \frac{u_2^2}{2g}$$

Physical meaning: In a steady, incompressible fluid, gravitational potential flow, the total specific energy (i.e., per unit weight) of each point of the ideal fluid is constant.

physical meaning

z	Specific potential energy
$\frac{p}{\rho g}$	Specific pressure energy
$\frac{u^2}{2g}$	Specific kinetic energy
$z + \frac{p}{\rho g}$	Total specific potential energy
$z + \frac{p}{\rho g} + \frac{u^2}{2g}$	Total specific energy

geometric meaning

Elevation head
Pressure head
Flow head
Hydraulic head
Total head





Chapter 3 Fundamentals of Fluid Motion

Section 5 Integration of Euler equations of motion

- 1, Integration for steady, potential, incompressible flow under only gravity
- 2, Path integration along streamline for steady, incompressible flow under only gravity





$$(f_x dx + f_y dy + f_z dz) - \frac{1}{\rho} \left(\frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz \right) = \frac{du_x}{dt} dx + \frac{du_y}{dt} dy + \frac{du_z}{dt} dz$$

<I> <II> <III>

2, Integral along the streamline

1, Steady, incompressible flow under only gravity

$$<I> = -gdz \quad <II> = \frac{1}{\rho} dp$$

2, In steady flow, streamline = pathline

$$\frac{dx}{dt} = u_x, \frac{dy}{dt} = u_y, \frac{dz}{dt} = u_z$$

$$<III> = u_x du_x + u_y du_y + u_z du_z = \frac{1}{2} d(u_x^2 + u_y^2 + u_z^2) = d\left(\frac{u^2}{2}\right)$$

• Energy equation along streamlines:

$$z + \frac{p}{\rho g} + \frac{u^2}{2g} = C$$

Note: The integral constant **C** is the same along the same streamline, but may differ for different streamlines for rotational flows.





1, What are the characteristics of real fluids compared with ideal fluids? How are the differential equations of motion of ideal fluids related to those of real fluids?

A real fluid is viscous and has shear stress; the left side of the equation in the differential equation of motion of the real fluid adds the term of shear stress due to viscosity to the differential equation of motion of the ideal fluid.

2, What kind of continuity equations are there? What does the continuity equation for incompressible fluids say?

General form, constant flow, incompressible flow; mass conservation (like for general flows)





3, What is the difference between the integral form of the Euler equations for potential flow and for flows along the streamline?

The mathematical expression is exactly the same, but the meaning is different.

The integral form for potential flow conditions is for any fluid particle in a steady, incompressible, potential flow of an ideal fluid, not limited to the same streamline. It doesn't work for rotational flows.

The integral along streamline form is for fluid particles at the same streamline in a steady, incompressible flow of an ideal fluid. It is applicable to rotational flows.

